Chapter 4: Graphs and transformations

Chapter 4 · Original slide 1 · Graphs and transformations

Pure Y1 Ch 4
Graphs and Transformations

Chapter 4 · Original slide 2 · starter

1) Factorise these quadratic expressions:

b) $x^2 + 6x + 5$

c) $x^2 - 4x + 3$

5) Sketch the graphs of the following functions:

f) $y = (x + 2)(x - 3)$

j) $y = x^2 - 6x - 7$

14) Copy and complete the table of values for the function $y = x^3 + x - 2$.

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
y	-12	-6.875			-2	-1.375			

Use the table of values to draw the graph of $y = x^3 + x - 2$.

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 3 · Chapter Overview

There are a few new bits and pieces since GCSE!

1a: Cubic Graphs

1b: Quartic Graphs

1c: Reciprocal Graphs

Sketch the graph with equation:
$y = x(x-3)^2$

Sketch the graph with equation:
$y = (x-1)^2(x+1)^2$

Sketch the graph with equation $y = -\frac{3}{x^2}$

NEW! to A Level 2017+

The old A Level only included cubic graphs, not quartics.

NEW! to A Level 2017+

In addition to graphs of the form $y = \frac{a}{x}$, you now need to recognise the sketch of $y = \frac{a}{x^2}$

Sketch the curves $y = \frac{4}{x^2}$ and $y = x^2(x-1)$ on the same axes. Using your sketch, state, with a reason, the number of real solutions to the equation $x^4(x-1) - 4 = 0$.

2: Points of Intersection

If $f(x) = x^2(x+1)$, sketch the graph of $y = f(x+a)$, indicating any intercepts with the axes.

3: Graph Transformations

NEW! since GCSE

The GCSE 2015+ syllabus included translations of graphs but not stretches.

Chapter 4 · Original slide 4 · Polynomial Graphs

In Chapter 2 we briefly saw that a polynomial expression is of the form:

$a + bx + cx^2 + dx^3 + ex^3 + \ldots$

where $a, b, c, d, e, \ldots$ are constants (which could be 0).

The order of a polynomial is its highest power.

Order	Name
0	
1	
2	
3	
4	
5	

These are covered in Chapter 5.

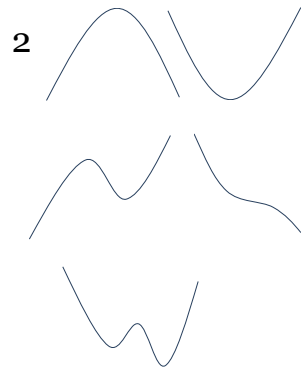
Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 5 · Polynomial Graphs



What property connects the order of the polynomial and the shape?

Order:

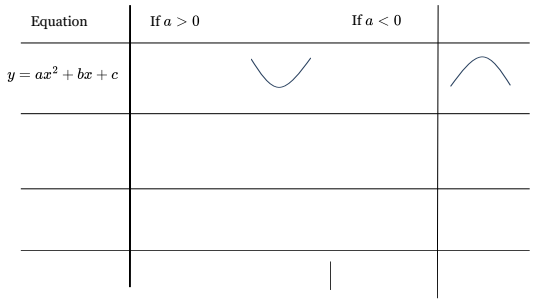
Bro Note: ...Actually this is not strictly true, e.g. consider $y = x^4$, which has a U shape. But this is because multiple turns are being squashed into a single point.

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

3

4

Chapter 4 · Original slide 6 · Polynomial Graphs



e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Resulting Shape

As $x \to \infty, y \to \infty$
As $x \to -\infty, y \to \infty$

$y = ax^3 + bx^2$
$+cx + d$

$y = ax^5 + bx^4 + \ldots$

(And we have the opposite if $a < 0$)

Resulting Shape

As $x \to \infty, y \to -\infty$
As $x \to -\infty, y \to -\infty$

$y = ax^4 + bx^3$
$+cx^2 + dx + e$

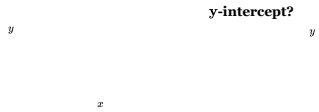
If a>0, what therefore can we say about the shape if:

• The order is odd:

• The order is even:

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 7 · Cubics





Sketch the curve with equation

$y = x^2(x-1)$

Shape?

Shape?

Roots?

y-intercept?

Sketch the curve with equation

$y = (x-2)(1-x)(1+x)$

Features you must consider:

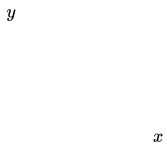
Tip: No need to expand out the whole thing. Just mentally consider the x terms multiplied together.

Roots?

This is sort of because the curve crosses at 0 then immediately crosses at 0 again!

Tip: It's incredibly easy to forget to write in one of the intercepts. So don't!

Chapter 4 · Original slide 8 · Cubics



Sketch the curve with equation

$y = (x-4)^3$

Shape?

Roots?

y-intercept?

Sketch the curve with equation

$y = (2-x)(x+1)^2$

Shape?

Roots?

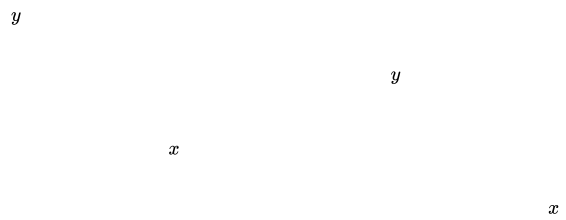
y-intercept?

Exam Notes: The term 'point of inflection' has been removed from the new A Level syllabus (bad idea!!).

A point of inflection is where the curve goes from 'convex' to 'concave' (or vice versa), i.e. curves in one direction before and curves in another direction after. You might have encountered these terms in Physics.

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 9 · Cubics with Limited Roots



Sketch the curve with equation

$y = (x + 1)\left(x^2 + x + 1\right)$

Shape?

Roots?

y-intercept?

However, in Chapter 12, we'll be able to work turning points using 'differentiation', and hence conclude that it doesn't have any!

Chapter 4 · Original slide 10 · Finding the equation yourself

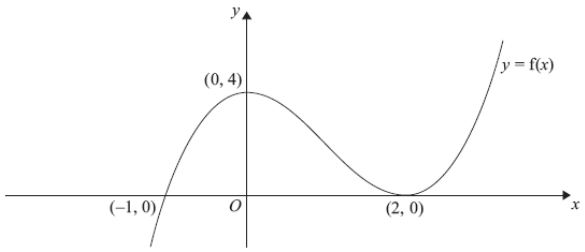


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.

The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).

The curve C has a maximum at the point (0, 4).

The equation of the curve C can be written in the form.

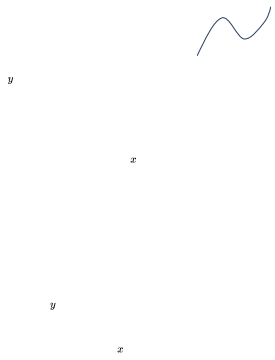
$y = x^3 + ax^2 + bx + c$

where a, b and c are integers.

(a) Calculate the values of a, b, c.

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 11 · Test Your Understanding



1

Sketch the curve with equation

$y = x(x-3)^2$

2

Sketch the curve with equation

$y = -(x+2)^3$

3

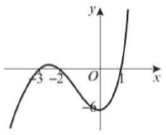
A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

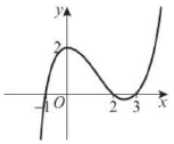
N

Sketch the curve with equation

$y = 2x^2(x-1)(x+1)^3$

Chapter 4 · Original slide 12 · Exercise 4A –Y1 Pure pages 63





5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b,c,d are real constants.

a Find the values of b, c, d. (3 marks)

b Write down the coordinates of the point where the curve crosses the y-axis. (1 mark)

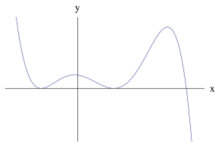
Chapter 4: Graphs and transformations

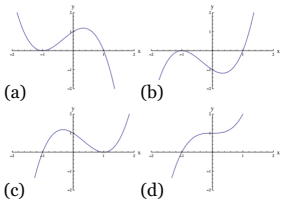
Chapter 4 · Original slide 13 · Exercise 4A –Y1 Pure pages 63

7 Given that $f(x) = (x - 10)(x^2 - 2x) + 12x$

a Express $f(x)$ in the form $x(ax^2 + bx + c)$ where a, b, c are real constants.
b Hence factorise $f(x)$ completely.
c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.

Chapter 4 · Original slide 14 · Exercise 4A Extension





Extension

2

[MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?

A) $y = (3 - x)^2(3 + x)^2(1 - x)$
B) $y = -x^2(x - 9)(x^2 - 3)$
C) $y = (x - 6)(x - 2)^2(x + 2)^2$
D) $y = (x^2 - 1)^2(3 - x)$

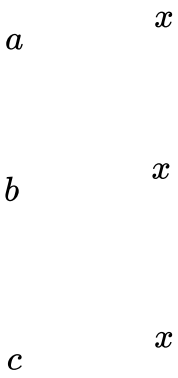
1

[MAT 2012 1E] Which one of the following equations could possibly have the graph given below?

Cubics can sometimes be factorised by pairing the terms:

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 15 · Recap



If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$:

Chapter 4 · Original slide 16 · Quartics



If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.

Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

Sketch the curve with equation

$y = x(x+1)(x-2)(x-3)$

Shape:

Roots:

y-intercept:

Sketch the curve with equation

$y = (x-2)^2(x+1)(3-x)$

Shape:

Roots:

y-intercept:

Chapter 4 · Original slide 17 · Quartics



Sketch the curve with equation
$y = (x+1)(x-1)^3$

Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a ‘touch’ point.

-1 root only appears once so line crosses at $x = -1$

+1 root triple repeated so point of inflection at $x = 1$

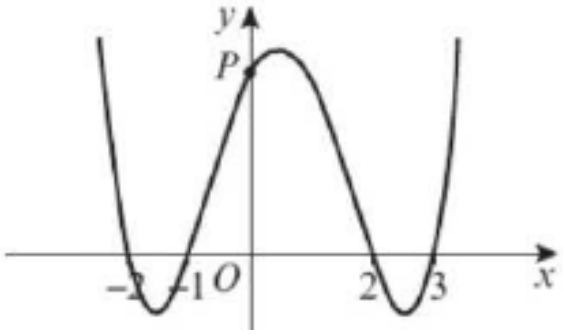
Chapter 4 · Original slide 18 · Test Your Understanding



Sketch the curve with equation
$y = x^2(x+1)(x-1)$

Sketch the curve with equation
$y = -(x+1)(x-3)^3$

Chapter 4 · Original slide 19 · Exercise 4B – Pure Y1 Pages 65-66



3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b,c,d,e are real constants.

a Find the coordinates of point P. (2 marks)

b Find the values of b, c, d, e. (3 marks)

4 Sketch the graph of
$y = (x + 5)(x - 4)(x^2 + 5x + 14)$.

Chapter 4 · Original slide 20 · Exercise 4B Extension

Extension

1

[STEP I 2012 Q2a]

a. Sketch $y = x^4 - 6x^2 + 9$

b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

By changing b, we shift the graph up and down.
Then we can see that:

i) 0 roots:

ii) 1 root:

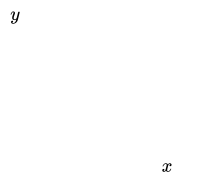
iii) 2 roots:

iv) 3 roots:

v) 4 roots:

b)

Chapter 4 · Original slide 21 · GCSE RECAP: Reciprocal Graphs



Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$

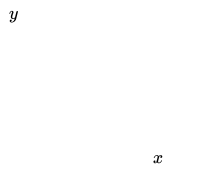
Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Notice the distance between this line and the x-axis (i.e. the line $y = 0$) gradually decreases as the lines go off towards infinity. The line $y = 0$ is known as an asymptote of the graph.

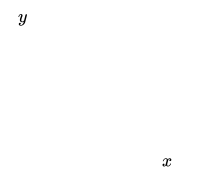
! An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$:

Chapter 4 · Original slide 22 · Reciprocal Graphs



This is new to the A Level 2017 syllabus.



Sketch $y = \frac{3}{x^2}$

Sketch $y = -\frac{4}{x^2}$

Hint: Note that anything squared will always be at least 0.

Chapter 4 · Original slide 23 · Reciprocal Graphs

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

$y = \frac{3}{x}$

$y = \frac{1}{x}$

y

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

x

Chapter 4 · Original slide 24 · Exercise 4C – Pure Y1 page 67

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$
d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Chapter 4 · Original slide 25 · Exercise 4C – Pure Y1 page 67

2 Use a separate diagram to sketch each pair of graphs.
a $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

Chapter 4 · Original slide 26 · Points of Intersection

In the previous chapter we saw why the points of intersection of two graphs gave the solutions to the simultaneous equations corresponding to these graphs.

If $y = f(x)$ and $y = g(x)$, then the x values of the points of intersection can be found when $f(x) = g(x)$.

Example: On the same diagram sketch the curves with equations $y = x(x - 3)$ and $y = x^2(1 - x)$. Find the coordinates of their points of intersection.

y

$\mathbf{y = x^2(1 - x)}$

$\mathbf{y = x(x - 3)}$

x

Tip: A classic mistake is to divide by x. NEVER divide an equation by a variable, because you lose a solution. Always factorise.

Chapter 4 · Original slide 27 · Further example involving unknown constants

On the same diagram sketch the curves with equations $y = x^2(3x - a)$ and $y = \frac{b}{x}$, where a, b are positive constants. State, giving a reason, the number of real solutions to the equation $x^2(3x - a) - \frac{b}{x} = 0$

$\mathbf{y = x^2(3x - a)}$

x

y

$\mathbf{y = \frac{b}{x}}$

Note: Note that the question is asking for the number of solutions, not the solutions themselves. We'd have to solve a quartic, with roots in terms of a and b. While there is a 'quartic formula' (like the quadratic formula), it is absolutely horrific.

Chapter 4 · Original slide 28 · Test Your Understanding

On the same diagram sketch the curves with equations $y = x(x - 4)$ and $y = x(x - 2)^2$, and hence find the coordinates of any points of intersection.

$\mathbf{y = x(x - 2)^2}$

$\mathbf{y = x(x - 4)}$

y

x

Hint: Remember you can use the discriminant to reason about the number of solutions of a quadratic.

Chapter 4 · Original slide 29 · Exercise 4D – Pure 1 Page 69-71

5 a On the same axes sketch the curves given by $y = x^2(x + a)$ and $y = \frac{b}{x}$ where a,b are both positive constants. (5 marks)
b Using your sketch, state, giving a reason, the number of real solutions to the equation $x^4 + ax^3 - b = 0$. (1 mark)

Chapter 4 · Original slide 30 · Exercise 4D – Pure 1 Page 69-71

6 a On the same set of axes sketch the graphs of $y = \frac{4}{x^2}$ and $y = 3x + 7$. (3 marks)

b Write down the number of real solutions to the equation $\frac{4}{x^2} = 3x + 7$. (1 mark)
c Show that you can rearrange the equation to give $(x + 1)(x + 2)(3x - 2) = 0$. (2 marks)
d Hence determine the exact coordinates of the points of intersection. (3 marks)

Chapter 4 · Original slide 31 · Exercise 4D – Pure 1 Page 69-71

10 a Sketch the graphs of $y = x^2 + 1$ and $2y = x - 1$.

b Explain why there are no real solutions to the equation $2x^2 - x + 3 = 0$.

c Work out the range of values of a such that the graphs of $y = x^2 + a$ and $2y = x - 1$ have two points of intersection.

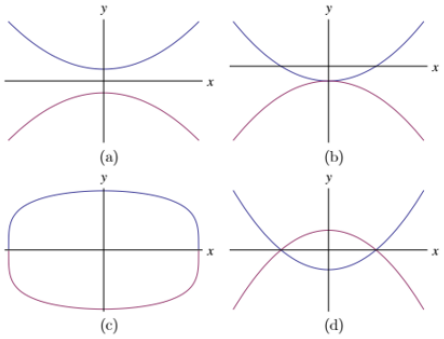
Chapter 4 · Original slide 32 · Exercise 4D – Pure 1 Page 69-71

11 a Sketch the graphs of $y = x^2(x - 1)(x + 1)$ and $y = \frac{1}{3}x^3 + 1$.

b Find the number of real solutions to the equation $3x^2(x - 1)(x + 1) = x^3 + 3$.

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 33 · Extension



1

[MAT 2005 1B]

The equation $\left(x^2+1\right)^{10} = 2x - x^2 - 2$

A) has $x = 2$ as a solution;

B) has no real solutions;

C) has an odd number of real solutions;

D) has twenty real solutions.

2

[MAT 2010 1A] The values of k for which the line $y = kx$ intersects the parabola $y = (x-1)^2$ are precisely

A) $k \leq 0$ B) $k \geq -4$

C) $k \geq 0$ or $k \leq -4$ D) $-4 \leq k \leq 0$

3

[MAT 2013 1D]

Which of the following sketches is a graph of $x^4 - y^2 = 2y + 1$?

Chapter 4 · Original slide 34 · Transformations of Functions

Suppose $f(x) = x^2$



Sketch $y = f(x)$:

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x+2)$?

Sketch $y = f(x+2)$

Chapter 4: Graphs and transformations

Chapter 4 · Original slide 35 · Transformations of Functions

This is all you need to remember when considering how transforming your function transforms your graph...

Affects which axis?

What we expect or opposite?

Change inside $f()$

Change outside $f()$

Therefore...

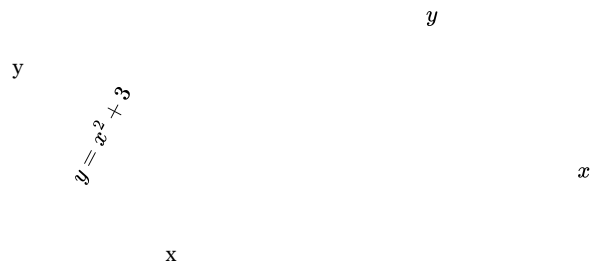
$y = f(x - 3)$

$y = f(x) + 4$

$y = f(5x)$

$y = 2f(x)$

Chapter 4 · Original slide 36 · Sketching transformed graphs



Sketch $y = \frac{2}{x+1}$

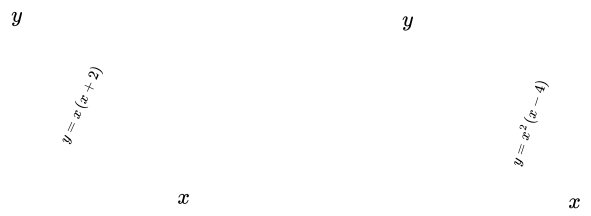
Sketch $y = x^2 + 3$

Imagine a sketch of $y = x^2$ and then do the translation, ensuring you adjust any intercepts with the axes.

The transformation might result in new intercepts or roots. You can find these in the usual way. Do not forget them!

Draw asymptotes using a dotted line and write its equation on it.

Chapter 4 · Original slide 37 · More Examples



Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$y = (2x)^2(2x-4)$

$y = (x-a)(x-a+2)$

$y = (2x)^2(2x-4)$

Note that our intercepts are in terms of a.

Chapter 4 · Original slide 38 · Reflections of Graphs

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

You did this at GCSE. The minus is outside the function, so affects the output, i.e. the y value.

The y values are negated, resulting in a reflection in the x-axis.

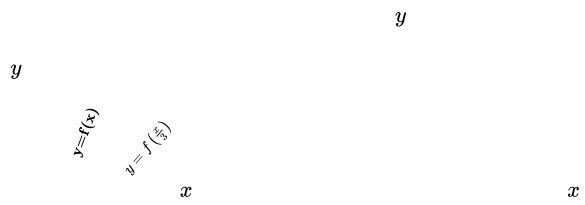
y

$y = f(x)$

$\mathbf{y = -f(x)}$

x

Chapter 4 · Original slide 39 · Test Your Understanding



Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

If $y = (x + 1)(x - 2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Chapter 4 · Original slide 40 · Exercise 4E – Pure Y1 page 74-75 (translations)

7 The point P(4,-1) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x - 2)$.
b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Chapter 4 · Original slide 41 · Exercise 4E – Pure Y1 page 74-75 (translations)

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$.
b The point with coordinates (-1,0) lies on the curve with equation $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find the two possible values of a.

Chapter 4 · Original slide 42 · Exercise 4E – Pure Y1 page 74-75 (translations)

12 a Sketch the graph of $y = x(x + 1)(x + 3)^2$.
b Find the possible values of b such that the point (2,0) lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$.

Chapter 4 · Original slide 43 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

7 The point P(2,-3) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(2x)$.
b State the coordinates that point P is transformed to on the curve with equation $y = 4\mathrm{f}(x)$.

8 The point Q(-2,8) lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.

Chapter 4 · Original slide 44 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

9 a Sketch the graph of $y = (x-2)(x-3)^2$.
b The graph of $y = (ax-2)(ax-3)^2$ passes through the point (1,0). Find two possible values for a.

Chapter 4 · Original slide 45 · Effect of transformation on specific points

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.

Where would each of these points end up?

$\mathbf{y=}f(x)$	$(\mathbf{4},\mathbf{3})$	$(\mathbf{1},\mathbf{0})$	$(\mathbf{6},\mathbf{-4})$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

Chapter 4 · Original slide 46 · Test Your Understanding

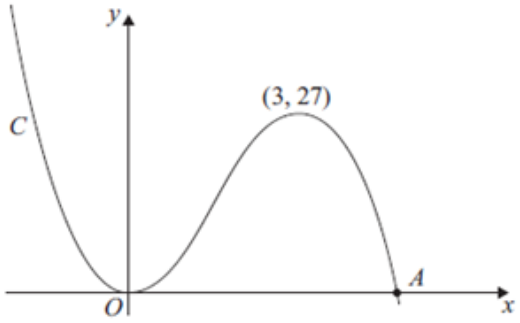


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point (3,27) and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)

(b) On separate diagrams sketch the curve with equation

(i) $y = \mathrm{f}(x + 3)$,

(ii) $y = \mathrm{f}(3x)$.

On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (6)

The curve with equation $y = \mathrm{f}(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.

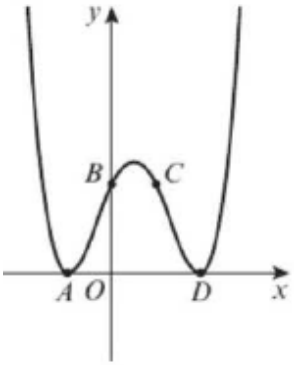
(c) Write down the value of k. (1)

Chapter 4 · Original slide 47 · Exercise 4G Pure Y1 Pages 80

5 The point P(2,1) lies on the graph with equation $y = \mathrm{f}(x)$.

a On the graph of $y = \mathrm{f}(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.
b Write down the coordinates of the point to which P maps under each transformation
i $\mathrm{f}(x - 4)$ **ii** $3\mathrm{f}(x)$ **iii** $\frac{1}{2}\mathrm{f}(x) - 4$

Chapter 4 · Original slide 48 · Exercise 4G Pure Y1 Pages 80



6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points A(-1,0),B(0,2),C(1,2),D(2,0) lie on the curve. Sketch the following graphs and give the coordinates of the points A,B,C,D after each transformation:

a $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$

e $2y - 1 = \mathrm{f}(x)$